

Animal armor

When sudden danger strikes, many animals rely on their speed to escape from trouble and make a dash for safety. Others have special body protection that helps them survive a surprise attack or defend themselves from hungry predators. Armadillos are protected by bony plates, but many other animals use armored shells, sharp spines, or thick hides to hold their enemies at bay. Horns and antlers can also fend off predators, but their main use is in ritual mating battles between rival males.

Armored ball

All species of armadillo have defensive armor made of interlocking bony plates, which protects their backs, heads, and parts of their legs and tail. If attacked, many armadillos try to run or dig their way to safety, but the three-banded armadillo stands its ground. This mammal can roll its flexible shell into a tight, impenetrable ball.

Prickly defenses

Spines and spikes make animals very difficult for predators to eat. Many animals have spines that stay upright all the time, but some are able to raise their spines as temporary protection when danger threatens.



Thorns

The thorny devil of Australia is a desert lizard covered with sharply pointed scales. In addition to being armored, it is well camouflaged against the desert sand.



Spikes

The porcupinefish raises its spikes by swallowing water, which makes its body swell up like a ball. When its body deflates, the fish's spines lie flat once more.



Spines

Sea urchins have chalky spines attached to raised knobs on their cases. The spines snap easily, embedding themselves in a predator's skin.



Prickles

A hedgehog's prickles are extra-thick hairs with sharp tips. If it is in danger, a hedgehog raises its spines and curls up into a tight, prickly ball.



Armor plating

The Indian rhinoceros is built like a living tank, with thick folds of armored skin. On its hind quarters and legs, the skin is reinforced with swellings that look like rivets, making it tougher still. Thanks to this armored suit, adult rhinos have no natural predators, but many fall victim to poachers, who hunt them for their horns.

Portable protection

Giant tortoises from the Galápagos Islands can weigh nearly one-third of a ton and have the biggest and heaviest shells of all land animals. Like all tortoises, they have soft shells when they first hatch, because the shell's bony plates are not joined together. As a tortoise grows, the plates start to fuse, giving the domed shell its tremendous strength. Most tortoises pull their heads and legs straight back into their shells to avoid danger. Long-necked tortoises bend their heads sideways to fit under their shells.

